

AP CSP | Lesson 26: Parallel & Distributed

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C defines sequential execution. A describes overlap; B is not required; D is an unrelated false guarantee.

Q2. Answer: A

A distinguishes overlap from device distribution. B wrongly excludes overlap; C ignores parallelism within one device; D ignores dependencies and serial work.

Q3. Answer: D

D: P1 handles 6+3 and P2 4+3, so $2+9=11$. A assumes all jobs start together; B halves total runtime; C assumes perfectly balanced workloads.

Q4. Answer: B

B is 18/8. A reverses the ratio; C subtracts; D adds runtimes. Speedup has no time unit.

Q5. Answer: A and D

A states the serial limit. D includes additional work. B ignores dependencies/load balance; C confuses computing capacity with data availability. Select A and D only.

Q6. Answer: C

C: $2+1+2=5$. A counts setup only; B omits overhead; D incorrectly duplicates work.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Job sum 16; runtime 18; jobs depend on setup; the jobs must be independent with processors available.

1 mark: Calculates job work 16.

1 mark: Calculates sequential runtime 18.

1 mark: States the setup dependency.

1 mark: Identifies job independence and/or available processing capacity for overlap.

Q8. 4 marks

Answer: C goes to P2 at time 6. D goes to P1 at time 8. P1 ends at 11. Total 11.

1 mark: Gives C to P2 starting at 6.

1 mark: Gives D to P1 starting at 8.

1 mark: Gives P1 finish time 11.

1 mark: Gives total runtime 11 as the latest completion.

Q9. 4 marks

Answer: 8; $18/8=2.25$; 14; four indivisible jobs already start together and the 6-unit job plus setup 2 remains.

1 mark: Gives runtime 8.

1 mark: Calculates speedup 2.25.

1 mark: Calculates overhead runtime 14.

1 mark: Explains the indivisible longest job/setup limit.

Q10. 4 marks

Answer: At least 2; at most 9; no, remaining work/coordination may still take time; indivisible tasks, limited processors, dependencies or overhead can limit more.

1 mark: Gives runtime lower bound 2.

1 mark: Gives speedup upper bound $18/2=9$.

1 mark: Explains that a bound does not promise attainment.

1 mark: Names a relevant further limitation.

Q11. 4 marks

Answer: ID3 is missing; retry/request it and verify receipt; processors cannot infer guaranteed missing data; multiple devices run the program.

1 mark: Identifies the absent required ID3.

1 mark: Describes recovery with receipt verification.

1 mark: Distinguishes missing input from processing capacity.

1 mark: Defines distribution across multiple devices.

Q12. 4 marks

Answer: It may tolerate a failed link when another working path remains. Endpoint isolation or packet loss may persist. Serial work, imbalance or overhead limits speedup. Specific failure and timing evidence is needed instead of universal guarantees.

1 mark: Links an alternative path to a specified tolerated link failure.

1 mark: Gives a valid residual network/end-to-end limit.

1 mark: Gives a valid dependency/serial/overhead performance limit.

1 mark: Rejects both universal claims using their missing conditions/evidence.